Literature Review and Research Gap

Inventory Management under Carbon Emission Regulations

and Time-Varying Demand

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Abstract

This document synthesises the findings of five seminal works that lie at the intersection of *inventory control*, *carbon-emission regulation*, and *time-varying demand*. The reviewed papers fall into two largely disjoint streams: (i) carbon-aware inventory models, in which classical Economic Order Quantity (EOQ) frameworks are extended to incorporate emission caps, carbon taxes, cap-and-trade markets, and green-technology investment; and (ii) inventory models with non-stationary demand patterns, particularly the *power demand pattern*. After critically appraising each contribution, the document identifies the gaps that remain at the confluence of the two streams and proposes specific directions for future research.

1 Introduction

The intensifying global focus on climate change has placed *operational* levers of carbon abatement at the centre of sustainable supply-chain research. Regulators across the European Union, North America, and Asia have implemented a variety of carbon-pricing mechanisms, including carbon taxes, mandatory emission caps, and cap-and-trade markets, while consumer pressure increasingly rewards firms that publicise green-technology investments and sustainability campaigns. In parallel, the classical inventory-control literature has relaxed the constant-demand assumption of the standard EOQ model and explored several time-varying demand structures, of which the *power demand pattern* captures both products whose demand intensifies near the end of the cycle (e.g., petrol, staple groceries) and products whose demand wanes (e.g., perishables).

Each of these two streams has matured independently, yet their intersection, namely *carbon-constrained inventory control under non-stationary, power-form demand*, remains comparatively unexplored. This review summarises five representative works that anchor each stream, exposes their methodological assumptions, and articulates the research gap that motivates further investigation.

2 Overview of the Reviewed Papers

Table 1 provides a high-level comparison of the five reviewed studies along the dimensions of demand structure, regulatory framework, decision variables, and central methodological contribution.

Table 1: Comparative overview of the five reviewed papers.

Reference	**Demand**	**Carbon Policy**	**Decision Variable(s)**	**Core Contribution**
Hua, Cheng & Wang (2011)[†]	Constant	Cap-and-trade	Order quantity	Closed-form EOQ extension under carbon trading; analyses effect of cap and price.
Benjaafar, Li & Daskin (2013)	Constant	Strict cap, tax, cap-and-offset, cap-and-trade	Lot size, multi-firm coordination	Lot-sizing models; quantifies value of operational adjustment vs. technology investment and supply-chain collaboration.
Hasan, Roy, Daryanto & Wee (2021)	Linear in green tech. & promotion	Carbon tax, cap-and-trade, strict cap	Order quantity, green-tech investment	Three integrated EOQ models with green investment and demand-stimulating promotion.
Sicilia, González-De-la-Rosa, Febles-Acosta & Alcaide-López-de-Pablo (2014)	Power pattern	None	Reorder point, scheduling period, lot size	Algorithmic optimal policy for production-inventory system with backlogged shortages and proportional production rate.

[†] The SSRN working paper (`ssrn1628953`) is the pre-print of Hua et al. (2011) and is therefore treated as a single contribution.

3 Thematic Literature Review

3.1 Stream 1: Carbon-aware Inventory Management

Hua, Cheng & Wang (2011)

Hua, Cheng and Wang formulate one of the earliest closed-form EOQ extensions in which a retailer operates under a cap-and-trade emission regime. The authors explicitly model carbon emissions arising from two operational sources: (i) transportation of replenishment orders, taken to be linear in the number of orders placed per unit time, and (ii) warehouse storage, taken to be linear in the average inventory level. By appending the purchased or sold carbon credits to the classical EOQ cost function, they derive an optimal order quantity Q^* and prove a clean structural result that compares Q^* to the cost-minimising EOQ quantity Q^0 and the emission-

minimising quantity $\hat{Q}$.

A central insight of the paper is the role played by the ratio g/e (warehouse-to-transport carbon intensity) versus the ratio h/K (warehouse-to-transport monetary intensity). When the two ratios coincide, the cost-minimising order quantity simultaneously minimises emissions, so participation in the carbon market is unnecessary; when they differ, Q^* falls strictly between Q^0 and $\hat{Q}$. The analytical and numerical results highlight that carbon trading *mechanically links* cost and emissions, but they rest on the strong assumption of *deterministic, time-invariant* demand and an exogenous carbon price.

Benjaafar, Li & Daskin (2013)

Benjaafar, Li and Daskin generalise the carbon-aware paradigm beyond a single firm and beyond cap-and-trade, building on the multi-period single- and multi-stage lot-sizing models that underlie much of the operations-management literature. Their contribution is fourfold. First, they show that emission parameters can be associated with each decision variable in classical lot-sizing problems, yielding tractable formulations under four distinct policies: strict caps, carbon tax, cap-and-offset, and cap-and-trade. Second, by varying these parameters numerically, they generate twelve managerial observations, of which the most striking is that *operational adjustment alone*, with no investment in greener technology, can deliver substantial emission reductions at modest cost. Third, they isolate the conditions under which tighter caps may paradoxically increase aggregate emissions, thereby informing policy design. Fourth, they evaluate the value of *supply-chain collaboration*, demonstrating that the gains depend sensitively on the prevailing regulatory mechanism and on whether emission caps are imposed firm-wise or supply-chain-wide.

The analytical core of the paper, an EOQ-style appendix that derives a piecewise-defined optimal order quantity under a strict cap, complements the multi-period programming results and unifies the single-firm treatment of Hua et al. (2011) with multi-firm extensions. Like its predecessor, however, the model presumes deterministic, stationary demand.

Hasan, Roy, Daryanto & Wee (2021)

The work of Hasan and colleagues represents the most recent and arguably the most behaviourally rich contribution to the carbon-aware EOQ literature reviewed here. They formulate three integrated profit-maximisation models that simultaneously optimise the order quantity Q and the green-technology investment G under (i) a carbon tax, (ii) cap-and-trade, and (iii) a strict carbon limit. Two features distinguish their formulation. First, the green investment is modelled by the concave function $R(G) = aG - bG^2$, capturing diminishing returns of abatement technology. Second, demand is modelled as $D(G, v) = D_0 + D_1 R(G) + mv$, so that demand is stimulated both by emission-reduction effort and by an explicit promotion level v.

Their numerical and sensitivity analyses produce three policy-relevant findings: (i) the carbon *tax rate* significantly drives total profit, whereas the cap level under cap-and-trade or under a strict limit is relatively benign; (ii) promotion is profitable across all three regimes; and

(iii) green-technology investment is profitable because it contracts the emissions exposure that must be priced through trading or limited by the cap. The model nevertheless remains rooted in a single buyer's EOQ structure with a *linear* demand function, i.e. no explicit time variation within a cycle.

3.2 Stream 2: Inventory Management under Power Demand

Sicilia, González-De-la-Rosa, Febles-Acosta & Alcaide-López-de-Pablo (2014)

The paper of Sicilia and colleagues anchors the second stream. They analyse a deterministic single-item production-inventory system in which demand follows the *power demand pattern* $D(t) = (r/T)\,(t/T)^{(1/n)-1}/n$, parameterised by the index $n > 0$. Values of n greater than one represent demand that intensifies as the cycle progresses (e.g., petrol or staple groceries running short toward the end of a replenishment cycle), while $n < 1$ captures demand that decays (e.g., perishable items losing appeal as the expiry date approaches). Production rate is taken to be proportional to demand rate, $P(t) = \alpha D(t)$ with $\alpha > 1$, so that inventory accumulates during the replenishment phase, after which a shortage-and-backlog phase concludes the cycle.

The authors minimise the cycle-averaged sum of holding, shortage, and ordering costs over two decision variables: the reorder point s and the scheduling period T. They prove that the first-order optimality condition reduces to a single non-linear equation in a normalised variable $x \in (0, 1-1/\alpha)$, demonstrate uniqueness of the root via the Intermediate Value Theorem, and propose a four-step algorithm to recover the economic lot size Q^* and the minimum cost C^*. Two limiting cases are recovered as sanity checks: (i) $n = 1$ collapses the model to the classical economic production quantity (EPQ) with backorders, and (ii) $\alpha \to \infty$ recovers the EOQ with power demand and instantaneous replenishment of Sicilia et al. (2012). The authors deliberately omit any environmental considerations, identifying deterioration and lost-sales settings as natural extensions, but *not* carbon emissions.

4 Synthesis and Cross-Cutting Observations

Three cross-cutting observations emerge from a comparative reading of the five papers.

First, demand is consistently simplified in carbon-aware models. Hua et al. (2011), Benjaafar et al. (2013), and Hasan et al. (2021) all retain a *constant-rate* demand assumption (or, at most, a static linear function of decision variables in Hasan et al.). None of these models allows demand to vary *within* an inventory cycle, even though much of the empirical inventory-management literature documents pronounced intra-cycle non-stationarity for many product categories. This simplification, while analytically convenient, may yield biased order-quantity recommendations whenever the underlying demand exhibits a strong power-form profile.

Second, the power-demand literature has remained environmentally agnostic. Sicilia et al. (2014), and the antecedent work they cite (Naddor, 1966; Datta and Pal, 1988; Lee and Wu, 2002; Sicilia et al., 2012; Mishra and Singh, 2013), focus exclusively on cost minimisation. Neither carbon-emission constraints nor green- technology investment have been embedded into the power-demand framework.

Third, regulatory variety is unevenly modelled across streams. The carbon-aware

stream has progressed from a single mechanism (cap-and-trade in Hua et al.) to multi-mechanism comparisons (Benjaafar et al. and Hasan et al., who jointly cover strict caps, taxes, cap-and-trade, and cap-and-offset). The power-demand stream offers no analogue. Hence, even if a power-demand inventory model were augmented with emissions, it would not be obvious *which* regulatory regime should accompany it.

5 Research Gap

Synthesising the above, we identify the following research gap, which motivates further investigation.

> *No existing study formulates an inventory model that simultaneously accommodates a power-form, time-varying demand pattern and one or more carbon-emission regulatory mechanisms (carbon tax, cap-and-trade, or strict cap), while also allowing for green-technology investment, backlogged shortages, and a finite production rate.*

This gap can be unpacked into the following more specific deficiencies.

G1. Demand realism in carbon-aware inventory models. The three carbon-aware studies reviewed assume constant or linear-in-investment demand. Extending the power-demand pattern of Sicilia et al. (2014) into the EOQ formulations of Hua et al. (2011) and Hasan et al. (2021) would (i) yield richer, more realistic order-quantity policies and (ii) test whether the central managerial insights of the carbon-aware literature, e.g., the dominance of operational adjustment over technology investment in Benjaafar et al. (2013), survive under non-stationary demand.

G2. Carbon footprint of shortage-and-backlog regimes. Sicilia et al. (2014) explicitly permit fully backlogged shortages, but no carbon-aware paper considers the emissions implications of backordering, e.g., expedited shipping or rush-mode transportation required to clear a backlog. A model that prices the *emissions cost of backlogged shortages*, in addition to the usual monetary shortage cost, would constitute a novel contribution.

G3. Joint optimisation of order quantity, scheduling period, and green-technology investment. The carbon-aware models reviewed optimise either order quantity (Hua et al.) or order quantity together with a scalar investment level (Hasan et al.); they do not jointly optimise an additional scheduling period, as is natural under a time-varying demand. Sicilia et al. (2014), by contrast, optimise the scheduling period but not the investment level. A unified model that treats Q, T, and G as co-decision variables remains absent.

G4. Regulatory comparison under non-stationary demand. Hasan et al. (2021) and Benjaafar et al. (2013) compare regulatory mechanisms only under static demand. Whether the qualitative ranking of mechanisms, e.g., Hasan et al.'s observation that the carbon tax rate matters more than the cap level, persists when demand follows a power pattern is an open empirical question with policy implications.

G5. Demand-investment coupling under power demand. Hasan et al. (2021) introduce

demand as an increasing function of green-technology investment and promotion level. No corresponding formulation exists in the power-demand literature; in particular, the *shape parameter* n of the power pattern itself may plausibly depend on green-marketing effort, an avenue that has been entirely overlooked.

G6. Deteriorating items at the intersection. Sicilia et al. (2014) explicitly flag deterioration as a future direction for the power-demand model, while none of the carbon-aware papers reviewed incorporates deterioration. A model that combines deterioration, power demand, and a carbon-pricing mechanism would address two open directions simultaneously.

Practical Implications. Closing this gap would benefit several stakeholders. Operations managers of products with strong power-form demand (e.g. fuel retailers, staple-food distributors, fresh-produce wholesalers) currently lack analytical tools that calibrate replenishment and shortage-management decisions to a carbon-priced environment. Policy makers, who already ask whether a carbon tax or a cap-and-trade scheme is preferable, would gain insight into how the answer is modulated by the demand profile of regulated industries. Finally, sustainability-oriented firms contemplating green-technology investment would acquire a sharper view of the conditions under which such investment yields net profit improvement.

6 Conclusion

The five papers reviewed here collectively map two well-developed but largely separate research streams. The carbon-aware EOQ stream, represented by Hua, Cheng & Wang (2011), Benjaafar, Li & Daskin (2013), and Hasan, Roy, Daryanto & Wee (2021), has progressively broadened the set of regulatory mechanisms, decision variables (now including green-technology investment), and behavioural ingredients (promotion-sensitive demand). The non-stationary-demand stream, exemplified by Sicilia, González-De-la-Rosa, Febles-Acosta & Alcaide-López-de-Pablo (2014), has refined the algorithmic toolkit for power-demand systems. Their integration, an inventory model that simultaneously features a power-form demand pattern, multiple carbon- emission regulatory mechanisms, optional green-technology investment, and backlogged shortages, constitutes a clear and as-yet-unaddressed research opportunity at the frontier of sustainable operations.

References

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